

Akshatha Dungi

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EXPERIENCE

IDEAS- Institute of Data Engineering, Analytics and Science Foundation

Jun. 2025 – Aug. 2025

Summer Research Intern

ISI Kolkata

- Engineered an Artificial Neural Network (ANN) achieving 97% multi-class classification accuracy, improving predictive performance by 15% over existing models
- MLOps:** Implemented evaluation metrics and visualization for stakeholder reporting
- Technical Stack: **Python, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib**

PROJECTS

AI Confidence-Calibration for Financial Time Series

- Led a 5-person research team to systematically evaluate calibration failure across 5 neural architectures (**MLP, Deep MLP, Vanilla RNN, LSTM, Residual MLP**) on **15 years of financial news stock data**
- Implemented Expected Calibration Error (**ECE**) tracking to measure confidence-accuracy misalignment over sequence lengths up to 100 timesteps
- Discovered that Vanilla RNNs fail earliest (**t=25**) but most predictably (**r=0.89 correlation with gradient norms**), while **LSTMs mask failures until sudden collapse (t=75, r=0.34)**
- Achieved **3.34% ECE** benchmark using Focal Calibration Loss, matching state-of-the-art results from **FinSen research**
- Built unified data pipeline processing 160K records** with custom loaders for all 5 models
- Published complete codebase with reproducibility guidelines on **GitHub**

Predictive Analysis of Iris Dataset

- Developed classification model to predict flower species based on morphological features
- Performed **exploratory data analysis** and feature correlation studies and created **data visualizations** to communicate model performance and insights.
- Built reproducible analysis pipeline using Python data science stack.

LEADERSHIP AND MANAGEMENT

a-hub, Incubation center

Jan. 2023 – May. 2023

Intern

Visakhapatnam

- Organized a university-wide internship drive, liaising with career services and 10+ corporate partners to **facilitate 30+ internship offers** across Business, Engineering, and IT fields.

EDUCATION

Andhra University College of Engineering, Visakhapatnam

Oct 2022 – Apr 2026

Bachelor of Technology in Computer Science and Systems Engineering

CGPA: 8.96/10

- Achievement:** Academic Excellence Award – Highest CGPA in CSE Department (2026)
- Coursework:** Data Structures, Algorithms, Databases, Computer Networks, Operating Systems, Object-Oriented Programming, Machine Learning

TECHNICAL SKILLS

Frontend : HTML5, CSS3, React, Responsive Design, JavaScript

Backend : Python(Flask), Java(Spring Boot), RESTful APIs

AI/ML Development : Python ML Stack(**scikit-learn, TensorFlow, PyTorch**), LLM Integration (OpenAI API, LangChain), AI-Assisted Development (GitHub Copilot), Neural Networks(**RNN, LSTM, MLP, Deep MLP, Residual Networks**), **Model Calibration, Expected Calibration Error (ECE), Reliability Diagrams, Time Series Analysis Forecasting**

Databases : MySQL, PostgreSQL, MongoDB

Cloud/DevOps : Microsoft Azure, Git, GitHub Actions, CI/CD Pipelines, Docker